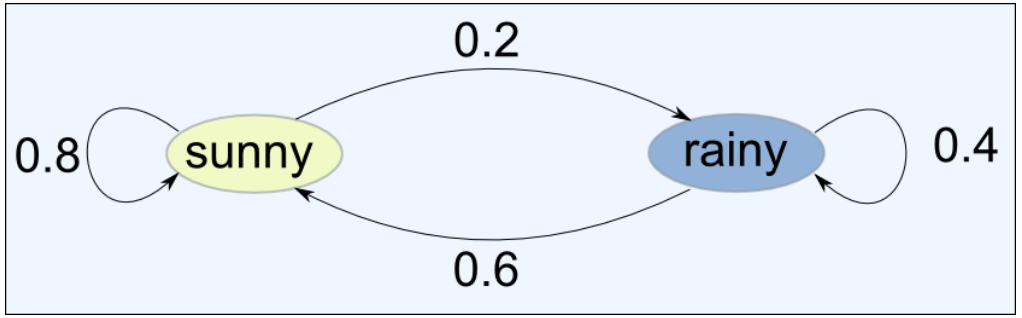
Hidden Markov Models

Markov Models

- A **Markov Model** (or Markov Chain) is a mathematical system that describes a sequence of possible events, where the probability of moving to any next state depends **only on the current state**, and not on the sequence of events that preceded it.
- This core assumption is known as the **Markov Property** (often described as "memoryless").

$$P(S_t \mid S_{t-1}, S_{t-2}, \ldots, S_1) = P(S_t \mid S_{t-1})$$

- This means that the current hidden state depends only on the immediately preceding hidden state, rather than the entire history.



- $P(\text{sunny} \mid \text{sunny}) = 0.8$ (Probability that tomorrow is sunny given today is sunny)
- $P(\text{rainy} \mid \text{sunny}) = 0.2$ (Probability that tomorrow is rainy given today is sunny)
- $P(\text{sunny} \mid \text{rainy}) = 0.6$ (Probability that tomorrow is sunny given today is rainy)
- $P(\text{rainy} \mid \text{rainy}) = 0.4$ (Probability that tomorrow is rainy given today is rainy)

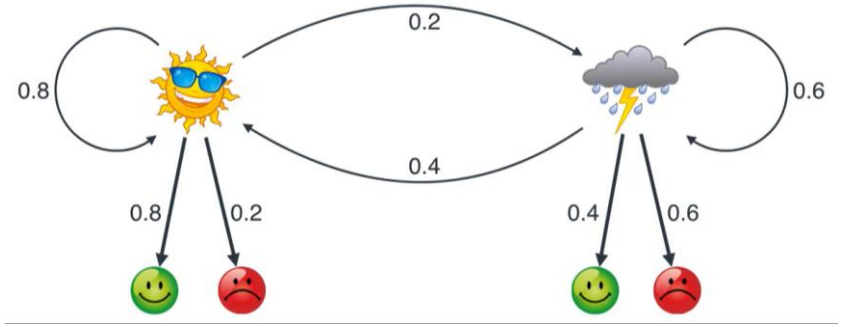
Hidden Markov Model (HMM)

- A **Hidden Markov Model (HMM)** is a statistical tool used to model systems that transition between different **hidden states**, but where we can only observe outputs that depend on those hidden states.
- The word **"Markov"** means the future depends only on the current state, not on the past.
- The word **"Hidden"** means you cannot directly see what state the system is in; you can only see clues (observations) that hint at it.
- HMM is widely used in:
  - Speech recognition
  - Natural language processing (NLP)
  - Bioinformatics (DNA sequence analysis)
  - Weather prediction
  - Time-series analysis

- An HMM consists of two types of variables:
  - Hidden states: The actual states of the system that cannot be directly observed.
  - Observations: The visible outputs generated by the hidden states.
- For example, consider predicting the weather based on whether a person carries an umbrella.
  - Imagine your friend works in a windowless room. Every day, they either **carry an umbrella** or **don't carry an umbrella**. You want to guess whether it is **Sunny** or **Rainy** outside based solely on whether they brought an umbrella.
  - Hidden state: Weather (Sunny or Rainy)
  - Observation: Umbrella (Present or Absent)
  - We cannot directly observe the weather in this example, but we can observe whether the person carries an umbrella.
- The HMM uses the observed sequence to estimate the most likely sequence of hidden states.
- An HMM is mathematically represented by:

$$\lambda = (A, B, \pi)$$

Component	Description
A	State transition probability matrix
B	Observation or emission probability matrix
π	Initial state probability distribution



A. State Transition Probability

It represents the probability of moving from one hidden state to another.

$$a_{ij} = P(S_t = j \mid S_{t-1} = i)$$

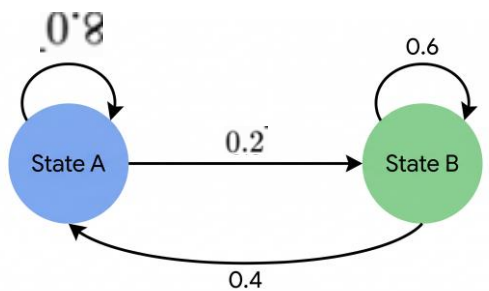
For example:

$$A = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}$$

Assume the states are Sunny and Rainy.

- $P(Sunny \rightarrow Sunny) = 0.8$
- $P(Sunny \rightarrow Rainy) = 0.2$
- $P(Rainy \rightarrow Sunny) = 0.4$
- $P(Rainy \rightarrow Rainy) = 0.6$

Each row sums to 1.



B. Observation (Emission) Probability

It represents the probability of observing an output given a hidden state.

$$b_j(o_t) = P(O_t = o_t \mid S_t = j)$$

For example:

Weather	Umbrella	No Umbrella
Sunny	0.1	0.9
Rainy	0.8	0.2

If the weather is rainy, the probability of observing an umbrella is 0.8.

C. Initial State Probability

It represents the probability of the system starting in a particular hidden state.

$$\pi_i = P(S_1 = i)$$

For example:

$$\pi = [0.6,\ 0.4]$$

This means that the initial probability of Sunny is 0.6 and Rainy is 0.4.

3. Markov Property

HMM is based on the Markov assumption:

$$P(S_t \mid S_{t-1}, S_{t-2}, \ldots, S_1) = P(S_t \mid S_{t-1})$$

This means that the current hidden state depends only on the immediately preceding hidden state, rather than the entire history.

Similarly, the observation at time t depends only on the current hidden state:

$$P(O_t \mid S_t, S_{t-1}, \ldots) = P(O_t \mid S_t)$$

These assumptions simplify the computation of probabilities.

- Given a sequence of observations, HMM estimates the most likely sequence of hidden states using the transition and emission probabilities.
- The joint probability of a hidden-state sequence and an observation sequence is:

$$P(S,O) = \pi_{s_1} b_{s_1}(o_1) \prod_{t=2}^{T} a_{s_{t-1}s_t} b_{s_t}(o_t)$$

- Three important algorithms are used in HMM:

1. Forward Algorithm: Computes the probability of an observed sequence.
2. Viterbi Algorithm: Determines the most likely sequence of hidden states.
3. Baum-Welch Algorithm: Estimates the model parameters from observed data using the Expectation-Maximization (EM) approach.

- **1. Forward algorithm (likelihood):** Given an HMM $\lambda = (A,B)$ and an observation sequence O, determine the likelihood $P(O|\lambda)$.
  - Ie, What is the probability of observing a particular sequence of observations?
  - For example, given the observation sequence:
    - O=(Umbrella, Umbrella)
    - Calculate: $P(O|\lambda)$
    - This is the probability of observing two consecutive umbrella observations under the given HMM.

- **2. Decoding Problem – Viterbi Algorithm:** Given an observation sequence O and an HMM λ =(A,B), **discover the best hidden state** sequence Q.
- Question: What is the most likely sequence of hidden states corresponding to the observations?
  - For example, suppose the observed sequence is:
  - O=(Umbrella,NoUmbrella,Umbrella)
  - The objective is to determine the most likely weather sequence:
  - S*=(Sunny,Rainy,Rainy)

- **3. Learning Problem – Baum-Welch Algorithm:** Given an observation sequence O and the set of states in the HMM, learn the HMM parameters A and B.
- Question: How can the transition and observation probabilities be estimated from training data?
  - Suppose we have many observations of umbrella usage, but we do not know the actual weather states.
  - The objective is to estimate:
  - lambda=(A,B,Π)
- The Baum-Welch algorithm uses the Expectation-Maximization (EM) approach to estimate the model parameters that maximize the likelihood of the observed sequence.

Summary

- HMM is a powerful probabilistic model for analyzing sequential data with hidden states. By modeling state transitions and observation probabilities, it can infer hidden sequences from observable data.
- Hidden Markov models are characterized by three fundamental problems:

Problem 1 (Likelihood):	Given an HMM $\lambda = (A, B)$ and an observation sequence O, determine the likelihood $P(O\|\lambda)$.
Problem 2 (Decoding):	Given an observation sequence O and an HMM $\lambda = (A, B)$, discover the best hidden state sequence Q.
Problem 3 (Learning):	Given an observation sequence O and the set of states in the HMM, learn the HMM parameters A and B.

- HMM uses observable data to infer hidden states. The Forward Algorithm evaluates an observation sequence, the Viterbi Algorithm identifies the most likely hidden-state sequence, and the Baum-Welch Algorithm learns the model parameters from data.